

Claude AI

Cheat Sheet

// prompting · models · capabilities · limits · best practices

claude.ai · 2026

● Current Models			
MODEL	API STRING	BEST FOR	SPEED
Claude Opus 4.8	claude-opus-4-8	Most capable, complex reasoning, frontier research	Slower
Claude Opus 4.7	claude-opus-4-7	Advanced tasks, high accuracy outputs	Moderate
Claude Opus 4.6	claude-opus-4-6	Deep analysis, nuanced long-form writing	Moderate
Claude Sonnet 4.6 ★	claude-sonnet-4-6	Everyday use — best speed/quality balance	Fast
Claude Haiku 4.5	claude-haiku-4-5-20251001	High-volume, low-latency, simple tasks	Fastest

● Prompt Anatomy		● Prompting Techniques	
ROLE You are a [role] with expertise in [domain].		Chain-of-thought	"Think step by step" before complex reasoning
CONTEXT Background: [paste relevant info here].		Few-shot	Give 2–3 input/output examples before your ask
TASK [Verb] a [output type] that achieves [core goal].		XML tags	Wrap:
FORMAT Respond in [format]. Max [N] words. Use [structure].		Role priming	"You are a senior..." sets tone and vocabulary
CONSTRAINTS Do NOT [thing to avoid]. Assume [audience / level].		Negative space	State what NOT to do: No bullet points, no code
		Output spec	"Under 200 words, for a 10-year-old, as JSON"
		Iterate	"Make punchier" / "Add example" / "Try again but..."
		Chain tasks	Outline first → draft each part → combine & refine

Prompt Starters, Capabilities & Features

● Prompt Starters – Copy & Adapt

SUMMARISE

Summarise the following in 3 bullet points for a non-technical executive:

[paste text here]

REWRITE

Rewrite the following to be [tone: concise / formal / friendly / persuasive].

```
Keep the meaning identical. Max [N] words: [paste
text]
```

STRUCTURED OUTPUT

Return your answer as a JSON array only – no markdown, no prose.

```
Each item: {"name": "", "summary": "", "score": 0}
```

DEBUG CODE

Here is my [language] code. It should [intended behaviour] but instead [actual behaviour].

Find and fix the bug: [paste code]

GENERATE OPTIONS

Give me 5 distinct approaches to [goal].

For each: a name, the core idea, pros, cons, and when to use it.

STEP-BY-STEP PLAN

Create a numbered action plan for [goal]. Each step: what to do, why it matters,

and any tools or resources needed. Audience: [level].

- **Capability Levels**

Task	GPT-4o (Green)	Claude 3.5 Sonnet (Blue)	Gemini 1.5 Pro (Orange)
Creative writing	95% (3 stars)	5% (1 star)	0%
Code gen & debug	90% (3 stars)	10% (2 stars)	0%
Reasoning	85% (3 stars)	15% (2 stars)	0%
Long documents	80% (3 stars)	20% (2 stars)	0%
Math (complex)	65% (2 stars)	35% (2 stars)	0%
Real-time info	60% (2 stars)	40% (1 star)	0%
Image gen	10% (0 stars)	90% (0 stars)	0% (X)

● Claude.ai Features

Web search	Real-time web access — toggle in conversation
Deep research	Multi-source synthesis across many searches
Artifacts	Preview standalone code, HTML, or docs inline
Projects	Persistent context: docs + custom instructions
Memory	Recalls facts across sessions (enable in Settings)
File uploads	PDFs, images, code — reads and reasons over them
Voice mode	Speak to Claude on mobile, hands-free
Code execution	Runs code and produces downloadable output files
Chrome agent	Browses the web autonomously (beta)
Claude Code	CLI tool for agentic coding in your terminal

Best Practices, Limits & API Reference

● Best Practices

■ Be specific

Name audience, format, length, and goal in the first message. Vague prompts return vague output.

■ Iterate in chat

Follow up: "make it shorter", "add a concrete example", "try a completely different angle".

■ Use XML structure

Wrap inputs in `<code>`, `</code>` tags to clearly separate them from your task.

■ Verify outputs

Cross-check all statistics, citations, code logic, and recent events. Claude can hallucinate confidently.

■ Paste real content

Don't describe what you have — give it. Claude reasons over actual text far better than descriptions of it.

■ Break tasks down

Outline → draft each part → combine. Multi-step beats one giant prompt almost every time.

■ Specify format

"JSON only", "markdown table", "numbered list", "plain prose" — Claude follows format instructions reliably.

■ Use Projects

Add reference docs and a system prompt to a Project so every chat shares the same context automatically.

● Hard Limits

CSAM / minors

Weapon synthesis

Malware / exploits

Cyberattack tools

Mass casualty help

Hate & incitement

Real person fiction

Self-harm facilitation

Non-negotiable — cannot be unlocked by any prompt, role, or framing.

● Watch-outs

Cutoff

Knowledge ends Aug 2025 — search for anything recent

Hallucinations

Confident-sounding wrong answers. Always verify facts.

No image gen

Claude cannot generate images (only understand them)

No memory*

Each chat starts fresh unless memory is enabled

Not a pro

Not a lawyer, doctor, or financial adviser — check output

Context limit

Very long chats may lose early context; start fresh

● API Quick Reference

<pre>// JavaScript – minimal call const res = await fetch('https://api.anthropic.com/v1/messages', { method: 'POST', headers: { 'x-api-key': YOUR_KEY, 'anthropic-version': '2023-06-01', 'content-type': 'application/json' }, body: JSON.stringify({ model: 'claude-sonnet-4-6', // or any model string above max_tokens: 1024, system: 'You are a helpful assistant.', // optional messages: [{ role: 'user', content: 'Hello!' }] }) }); const data = await res.json(); console.log(data.content[0].text);</pre>	stream	"stream": true → token-by-token SSE streaming response
	tools	"tools": [{name, description, input_schema}] for function calling
	vision	messages content: [{type:"image", source:{type:"base64",...}}]
	temp	"temperature": 0.0–1.0 (default 1.0, lower = more deterministic)
	top_p	"top_p": nucleus sampling alternative to temperature
	system	Top-level "system" key for instructions outside the message turn
	multi-turn	Append {role:"assistant", content:"..." } to messages for history
	batch	POST /v1/messages/batches for async bulk processing (cheaper)